

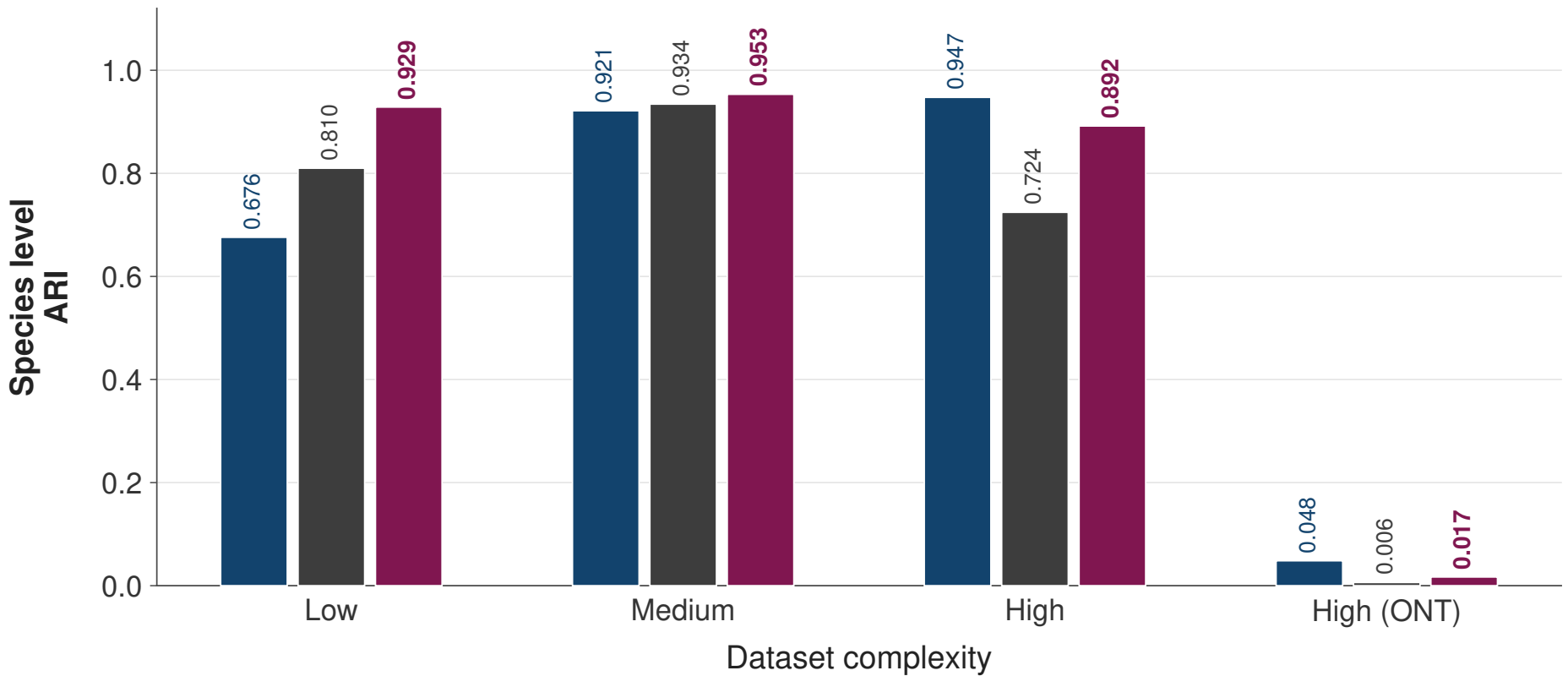
Simulated benchmark: clustering agreement and bin recovery (metric-specific top 3)

Methods appearing in metric-specific top 3

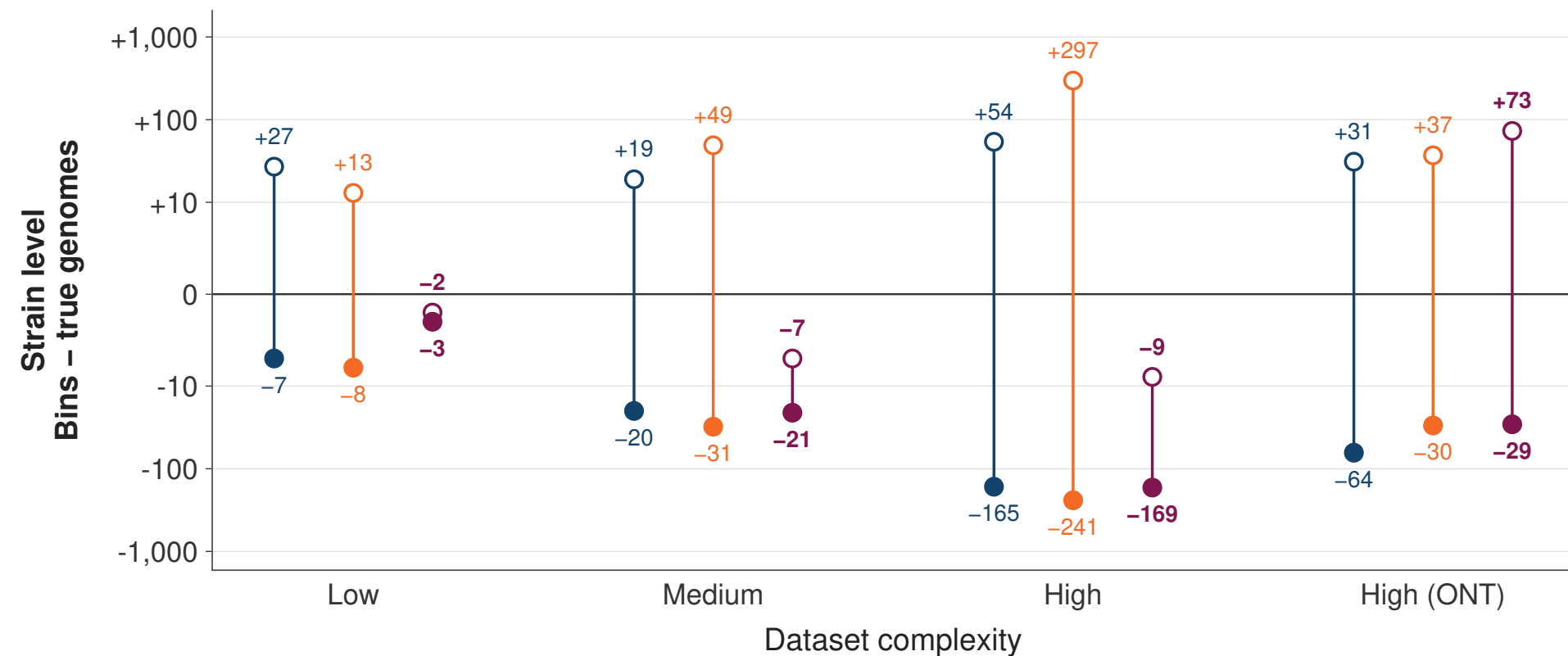
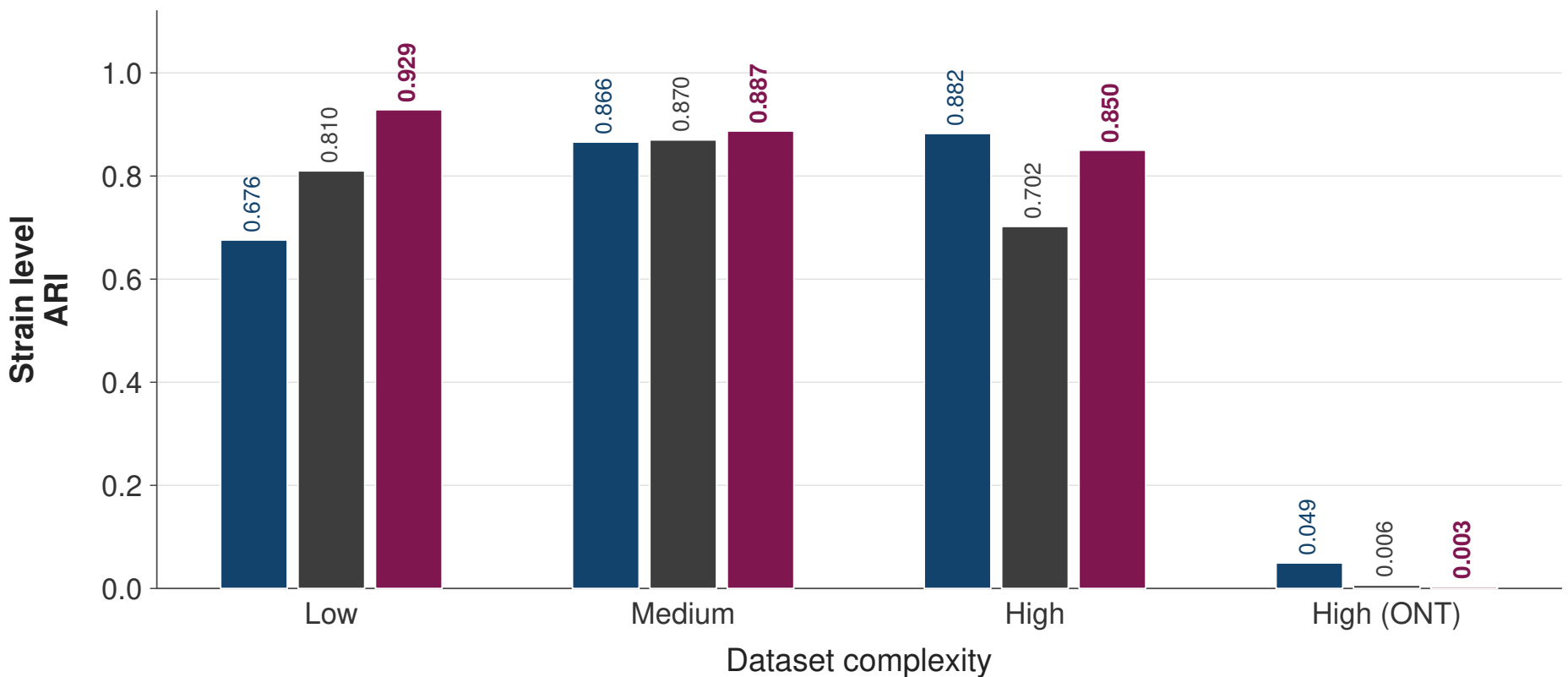
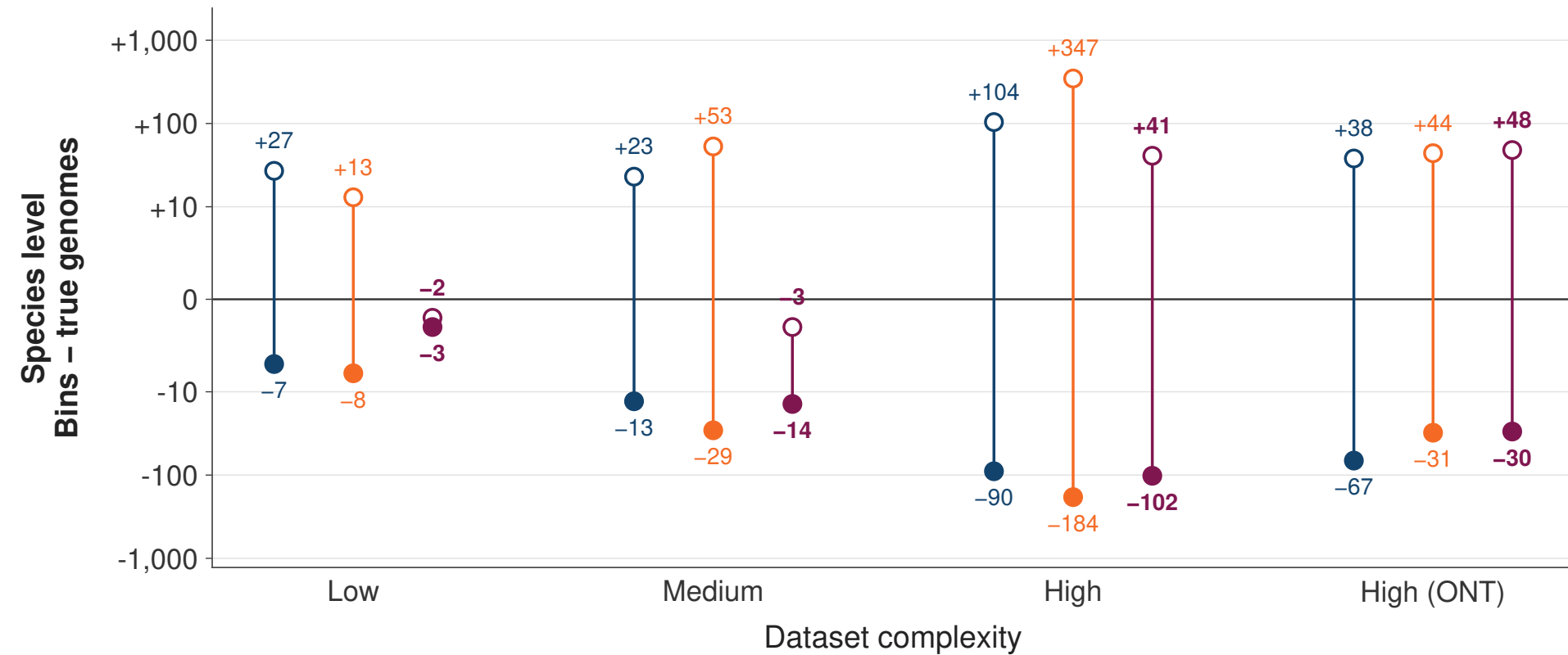
■ VAMB ■ vRhyme ■ MetaBAT 2 ■ MAGE-Bin

○ Predicted bins – true genomes ● HQ bins – true genomes — Reference line at 0

(a) Adjusted Rand index



(b) Bins relative to true genomes



Top 3 selected separately per panel using mean performance across the four simulated datasets and both levels (ARI: highest; bins: smallest |HQ bins – true genomes|). Panel (b) uses a symmetric-log y-axis, linear within ± 10 .